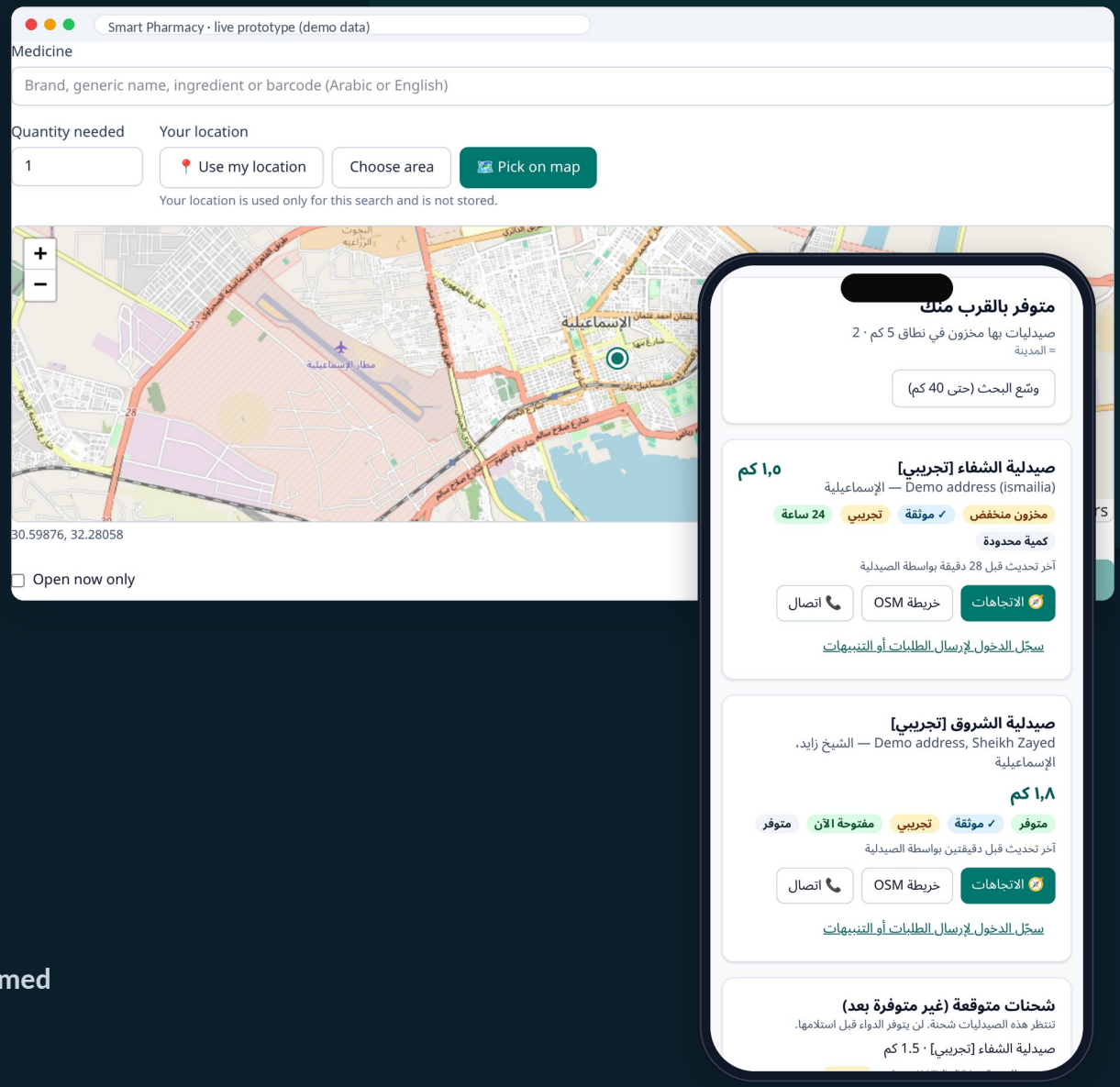
HACKATHON PITCH · 2026

Smart Pharmacy

الصيدلية الذكية

Find the nearest pharmacy that actually has your medicine — verified, fresh, in Arabic & English.

Patients | Pharmacies | Warehouses



Live prototype · demo data · Ismailia, Egypt

Ahmed Alsaholy · Ahmed Shahen · Khaled Zakarya · Mohamed Ekramy · Fayez Mohamed

THE PROBLEM

When a medicine disappears, patients go pharmacy to pharmacy

ILLUSTRATIVE SCENARIO

Um Ahmed, 58, lives in Ismailia and takes Concor 5 mg for her blood pressure.

Her pharmacy is out. She calls six pharmacies, visits three — and still can't tell who has it, or when the next delivery arrives.

Concor, Augmentin, Brufen and Cataflam were all reported short in 2024 [5] [11].

Patients

Lost time and money; some turn to black-market packs of unknown origin [11].

Pharmacists

About 30% of their time goes to checking stock and calling suppliers by hand [12].

Warehouses

No live view of where demand is — orders arrive by phone, shortages are spotted late.

~1,000

of 17,000 registered medicines hit by shortages (Jul 2024) [5]

30M+

Egyptians with hypertension depend on chronic medicines [5]

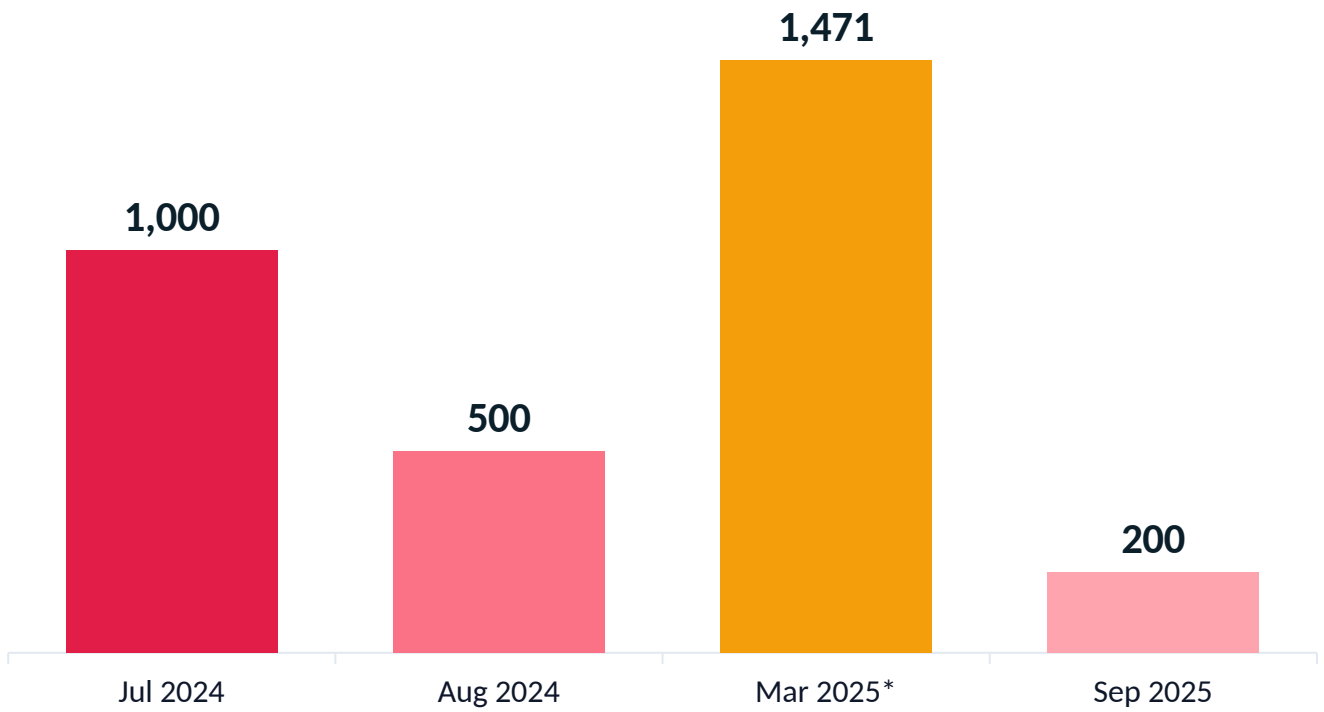
11M

people live with diabetes and need insulin and tablets every month [11]

THE PROBLEM IN NUMBERS

Shortages come and go — and nobody shares one live picture

Reported number of medicines in shortage — Egypt



** Branded trade names (FEI Pharmaceutical Chamber). Counts come from different bodies and definitions — that inconsistency is part of the problem [5][6][8][9].*

20–40%

of trade names were short in mid-2024, while 15–20% of active-ingredient demand went unmet [7].

300

branded drugs had no available alternative in March 2025 [8].

40% → <5%

shortage share by late 2025 — yet insulin, cardiac and oncology medicines still go missing [9][10].

Patients need to know where the last boxes are — today.

ROOT CAUSE

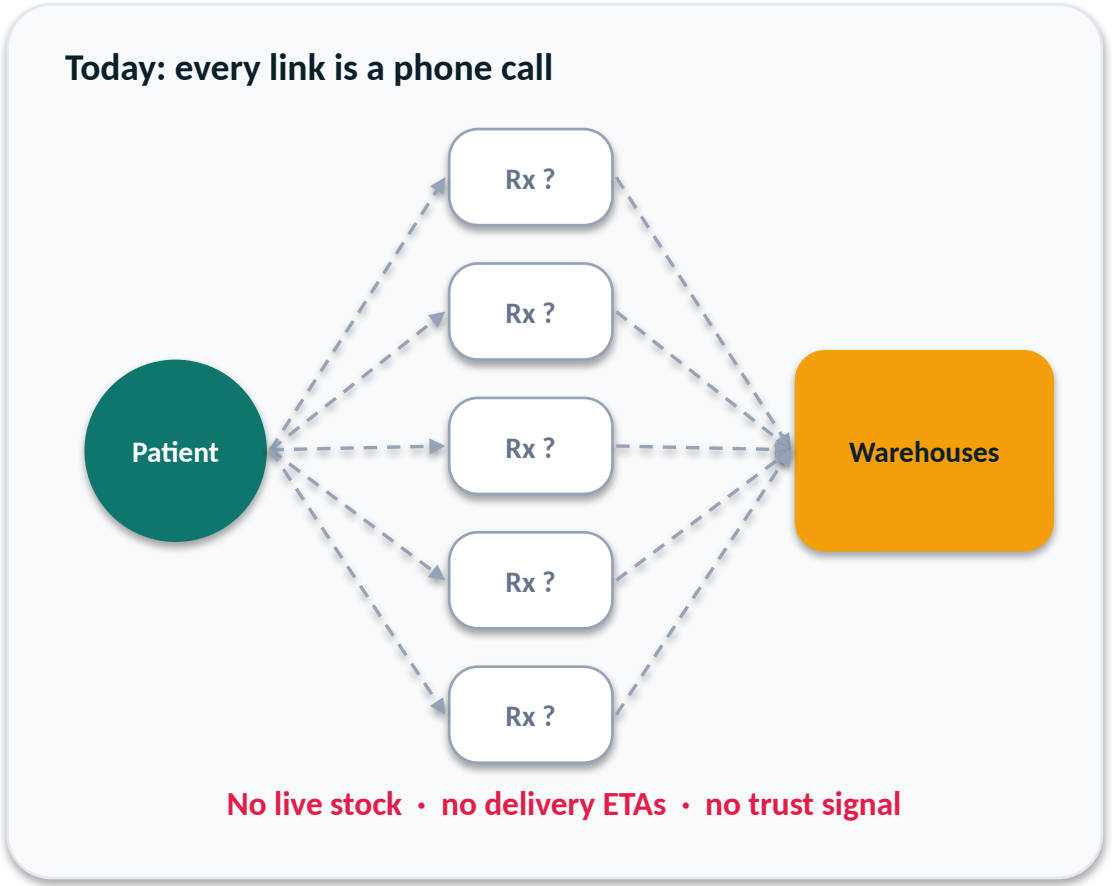
95,000 pharmacies. Zero shared stock visibility.

95,000
community pharmacies — one of the highest densities in the world [2]

≤ 2
pharmacies per pharmacist by law → mostly independent shops, no chain-wide inventory [13]

7,000+
products from ~3,000 suppliers, still ordered largely by phone [12]

~30%
of a pharmacist's time spent checking stock and prices manually [12]

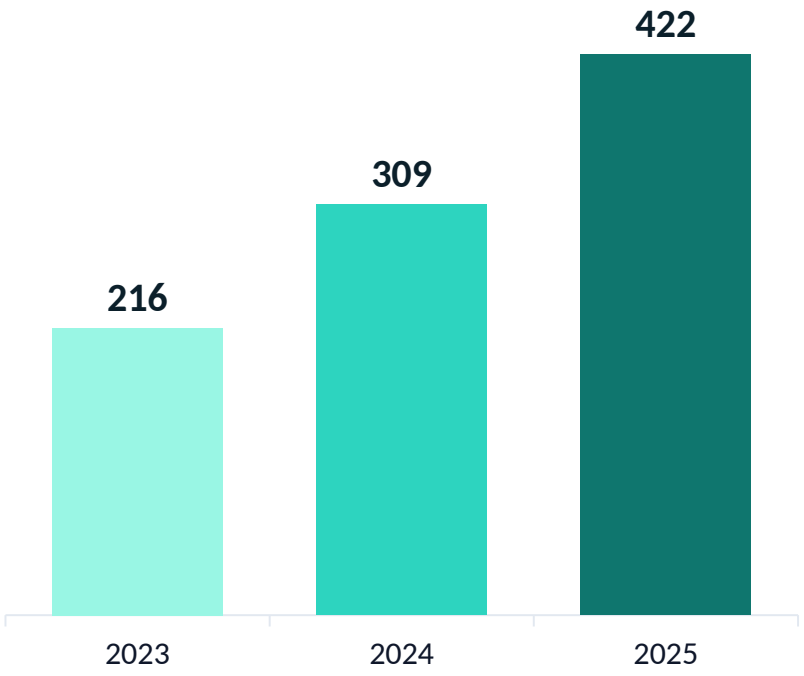


Sources: [2] [12] [13]

OPPORTUNITY

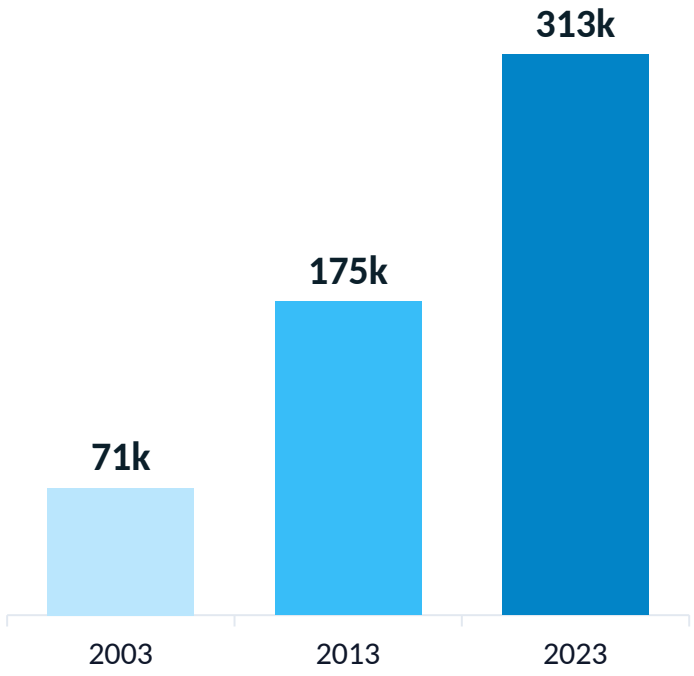
A large, fast-growing and digital-ready market

Egypt pharma market (EGP bn)



+43% in 2024 · +37% in 2025 — Egyptian Drug Authority [3][4]

Registered pharmacists



4.4× in 20 years — Pharmacists Syndicate records [2]

109.5M
people in Egypt (CAPMAS, Sep 2026) [1]

96.3M
internet users · 81.9% penetration (Jan 2025) [14]

92.6M
mobile-internet subscriptions (Dec 2025) [15]

~4B
medicine packs sold in 2025 (EDA) [4]

Sources: [1]–[4] [14] [15]

OUR SOLUTION

One verified network for medicine availability

Smart Pharmacy connects patients, pharmacies and warehouses around one question: who has this exact medicine, right now, near me?

For patients

- Search by brand, generic, ingredient or barcode — Arabic or English
- Nearest verified pharmacies with fresh stock, 5 → 10 → 20 → 40 km
- Directions, call, ask to confirm or hold — or get an alert

For pharmacies

- Inventory with batches, expiry and low-stock alerts
- Every unit change recorded in an auditable ledger
- Order from warehouses; track incoming stock and ETAs

For warehouses

- Accept or partially fill orders with automatic allocation
- Shipments, delivery dates and delays — with reasons
- Resolve discrepancies; see real demand signals

Trust layer: admin-verified pharmacies · freshness labels · auditable stock ledger · privacy by design

HOW IT WORKS

From search to pickup — and from warehouse to shelf

1 Search

Exact product: brand, generic, ingredient or barcode — Arabic or English

2 Locate

GPS (with consent), city / district, or a pin on the map

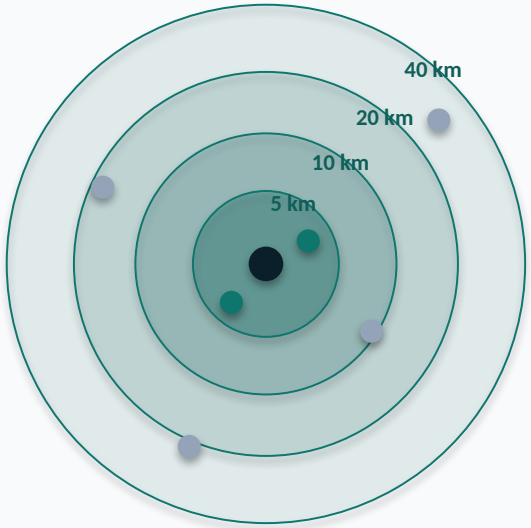
3 Rank

Verified & fresh stock first; expand 5 → 10 → 20 → 40 km

4 Act

Directions, call, confirm or hold — or set an alert

Progressive radius search

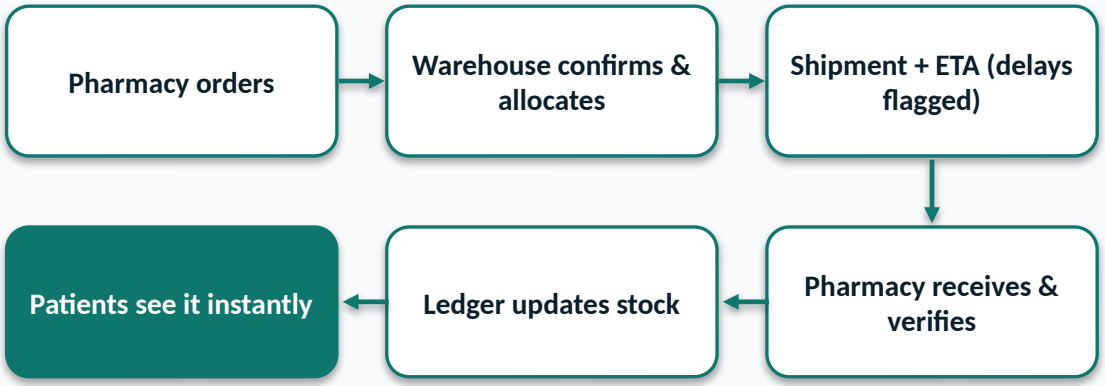


Stops at the first ring with verified, fresh stock.

Patients can expand anytime — up to 40 km (configurable).

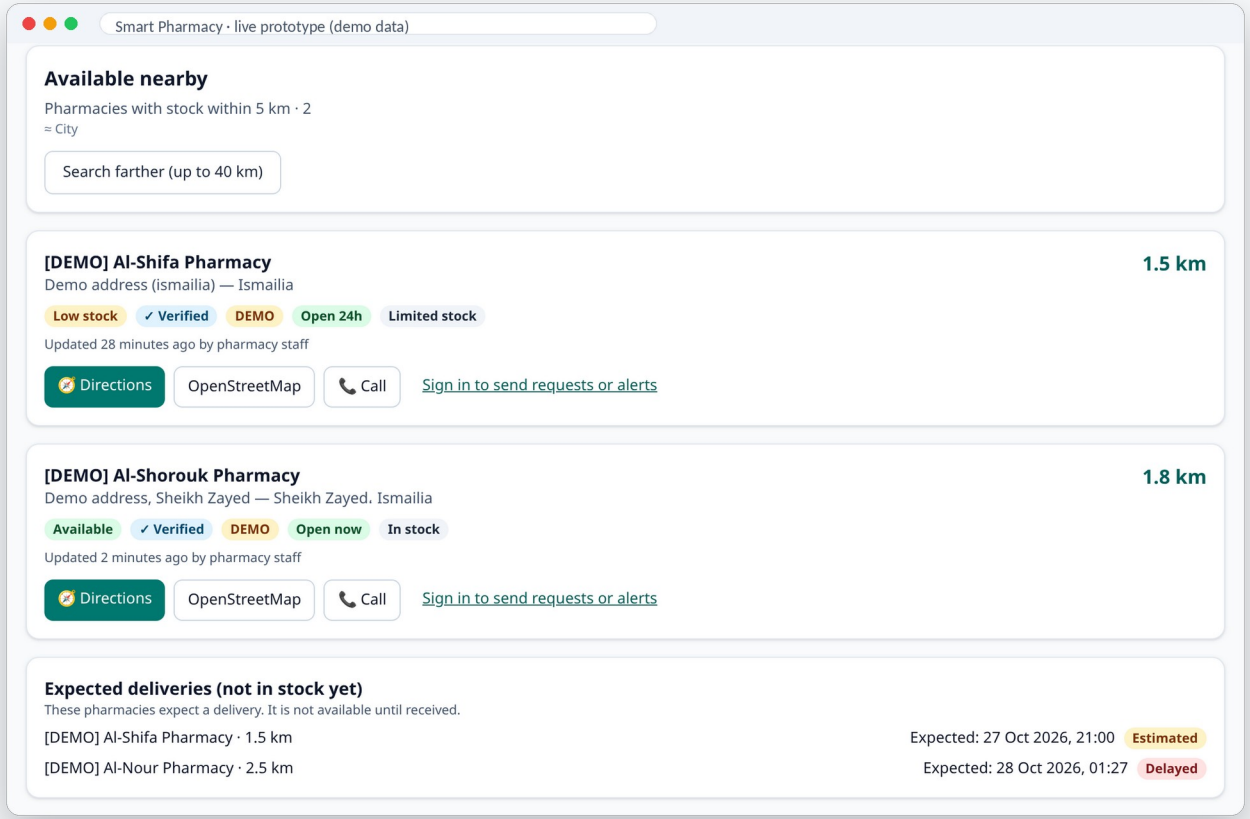
One indexed query; results in milliseconds.

Supply loop keeps stock honest



Orders and dispatches never inflate pharmacy stock — only a verified receipt does.

Patient finder — exact product, nearest verified stock



Distance · verified · open now · freshness — expected deliveries are kept separate, never counted as stock



Exact-product picker (5 mg ≠ 2.5 mg) · Arabic-first, right-to-left, no account needed

Pharmacy workspace — inventory you can trust

PRODUCT	PHYSICAL	RESERVED	UNAVAILABLE	AVAILABLE	STATUS	UPDATED	
Augmentin 625 mg film_coated_tablet · 14 tablet	12	1	0	**11**	AVAILABLE	33 minutes ago MANUAL	Manage
Concor 5 mg film_coated_tablet · 30 tablet	3	0	0	**3**	LOW_STOCK	33 minutes ago MANUAL	Manage
Panadol 500 mg film_coated_tablet · 24 tablet	33	0	1	**32**	AVAILABLE	7 minutes ago MANUAL	Manage

‹ 1 / 1 ›

Incoming stock is NOT part of your available inventory until you receive and verify it.

WAREHOUSE	ORDER	MEDICINE	REQ / CONF / ALLOC	DISPATCHED	RECEIVED	SHIPMENT	ETA	LAST SCHEDULE UPDATE
[DEMO] Suez Canal Medical Distribution	PO-2026-000001 IN_FULFILLMENT	Concor 5 mg	30 / 30 / 0	30 (26 Oct 2026, 19:00)	0	SH-2026-000001 DISPATCHED	27 Oct 2026, 21:00 ESTIMATED	33 minutes ago

Panadol 500 mg AVAILABLE ×

Physical **33** − Reserved **0** − Unavailable **1** = Available **32**

Change type: STOCK_COUNT | Counted qty: 0 | Note | Apply | Still accurate

Movement history (append-only ledger)

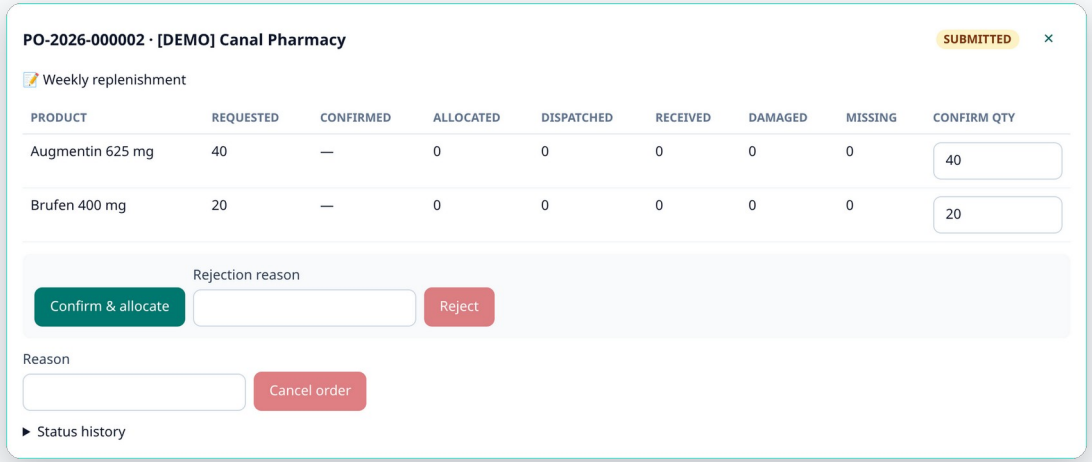
WHEN	TYPE	Δ PHYS / RES / UNAV	AFTER	BY
26 Oct 2026, 19:27	STOCK_COUNT · PHYSICAL_COUNT Evening count	-2 / 0 / 0	33 / 0 / 1	Al-Shifa Pharmacy Owner
26 Oct 2026, 19:27	MARK_UNAVAILABLE · DAMAGED Crushed box	0 / 0 / 1	35 / 0 / 1	Al-Shifa Pharmacy Owner
26 Oct 2026, 19:27	SALE · SALE	-2 / 0 / 0	35 / 0 / 0	Al-Shifa Pharmacy Owner
26 Oct 2026, 19:27	SALE · SALE	-3 / 0 / 0	37 / 0 / 0	Al-Shifa Pharmacy Owner
26 Oct 2026, 19:00	OPENING_BALANCE Opening balance	40 / 0 / 0	40 / 0 / 0	Al-Shifa Pharmacy Owner

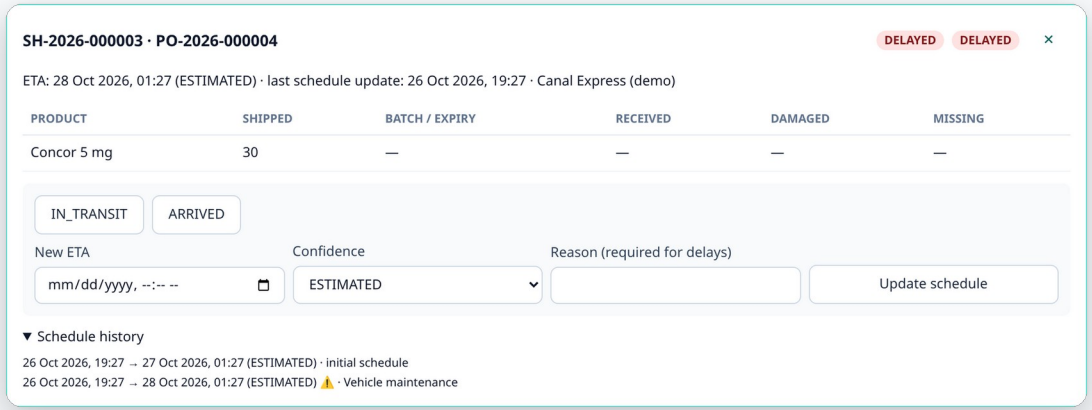
- Low-stock & expiry alerts · CSV import · barcode lookup
- Incoming stock tracked from order to verified receipt — separate from shelf stock
- Patient confirmation requests and holds with pickup codes

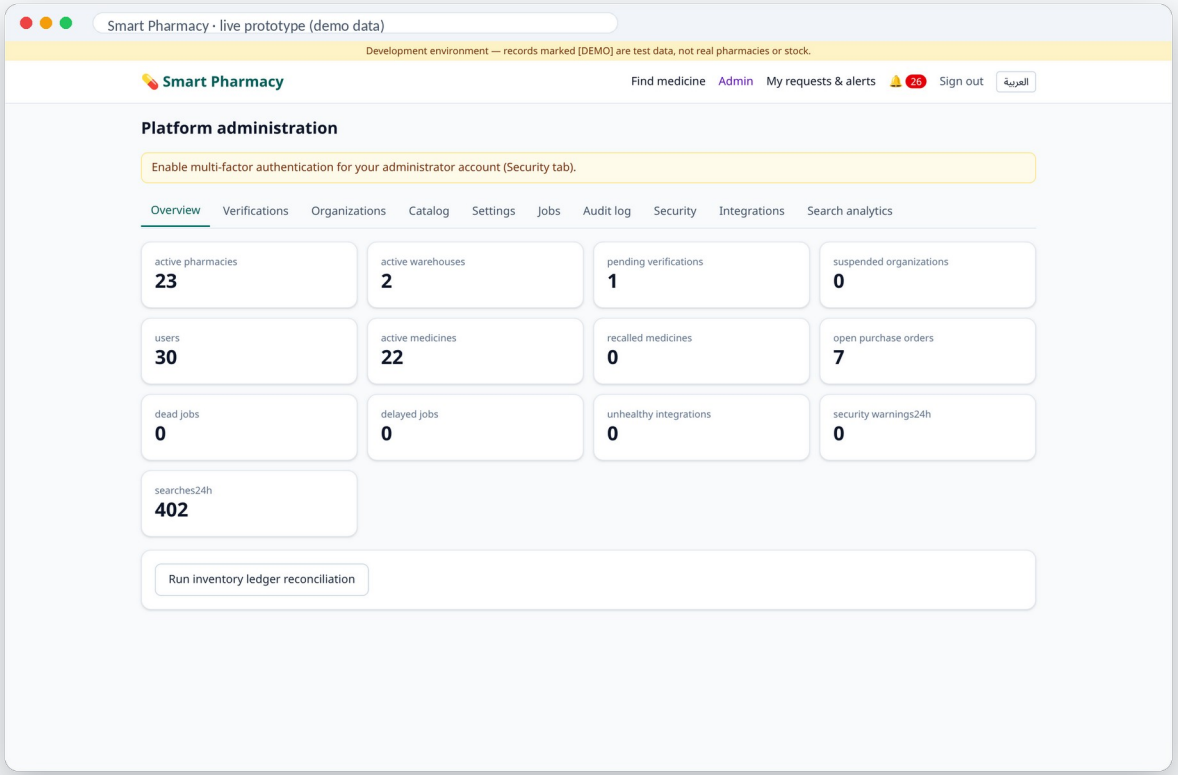
Available = Physical − Reserved − Unavailable

Never negative — enforced by the database. Every change is an append-only ledger entry.

Warehouse operations and platform trust controls







- Confirming an order reserves stock instantly — no overselling
- Delays require a reason; pharmacies are notified automatically
- Admins verify every pharmacy before patients can see it

TRUST BY DESIGN

What patients see is verified, fresh — and provable

Freshness windows (configurable)

Update source	Confirmed	Outdated after
Pharmacy POS integration	2 h	24 h
Staff update · barcode · delivery receipt	12 h	72 h
CSV import	24 h	7 days

A failing or silent integration is automatically downgraded — it can never look fresh.

What patients see

- Available
- Low stock
- Recently reported — unconfirmed
- Fewer units than requested
- Outdated — call first
- Expected delivery — not in stock

Guarantees

- Available stock can't go negative (DB-enforced)
- Append-only ledger: every unit traceable
- Duplicate-safe orders & syncs
- Pharmacies can't touch each other's data
- Admin MFA, audit logs, rate limits
- Location used once, never stored

Ranking rule: a closer but unconfirmed listing never outranks a farther verified and recently updated one.

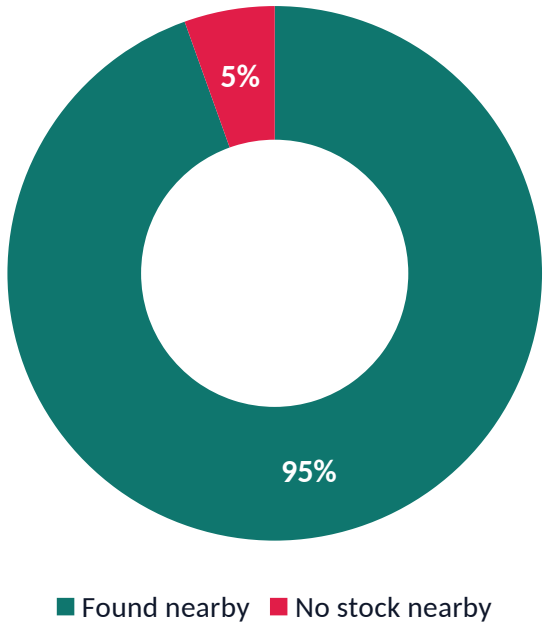
Sources: [18]

DATA FROM THE PROTOTYPE

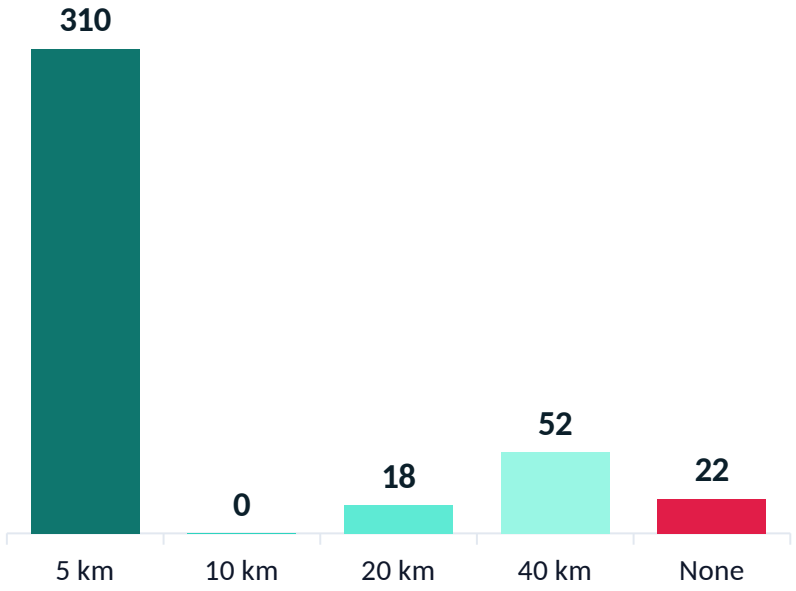
Every search becomes a signal

Simulated traffic: 402 patient searches across 24 demo pharmacies in Ismailia governorate — demo data, shown to illustrate the analytics, not market results.

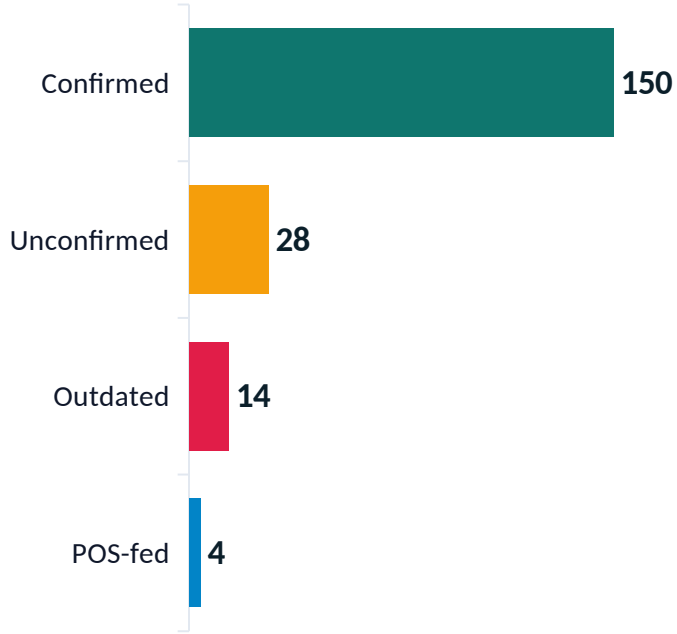
Search outcomes



Where stock was found



Freshness of listings



Insight: all 22 unmet searches were for Mixtard 30 100 IU/mL — stocked only at a warehouse. That is the demand signal suppliers and authorities need, captured without storing personal data.

Sources: [18] — live query of the prototype database

PROOF IT WORKS

A working prototype — measured, not mocked

67/67
automated tests passing

20/20
end-to-end acceptance steps

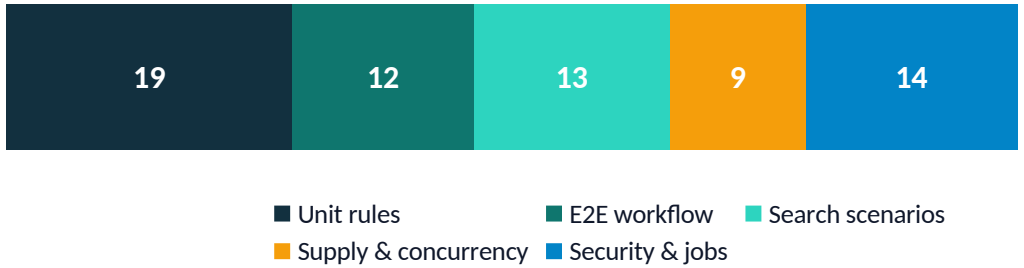
95
documented REST API endpoints

0
stock-ledger mismatches

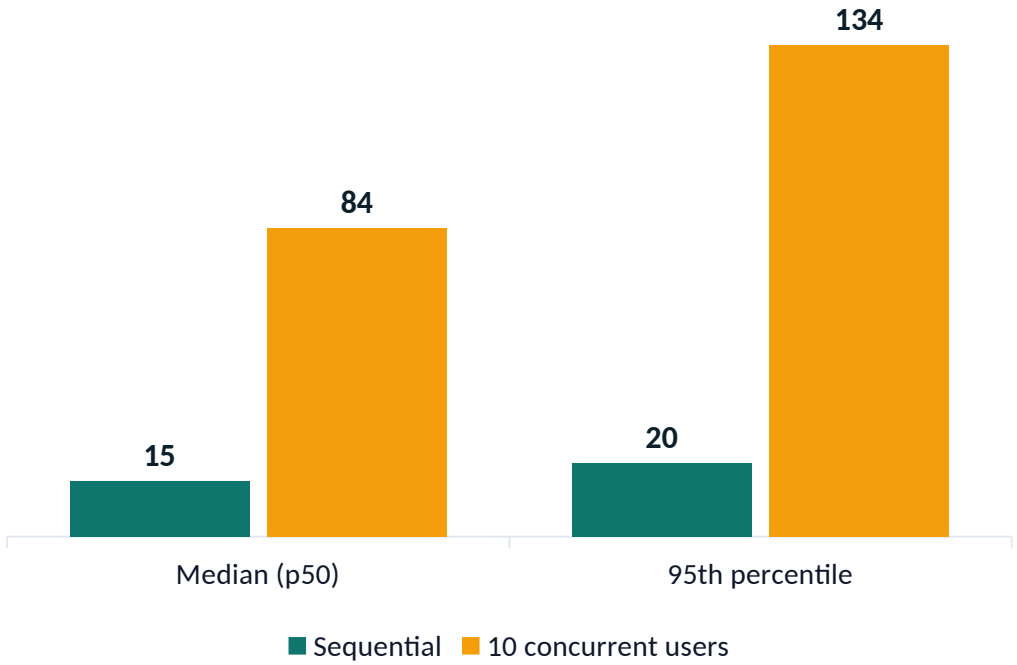
0
known dependency vulnerabilities

AR · EN
full right-to-left support

Automated tests by area



Finder response time (ms)

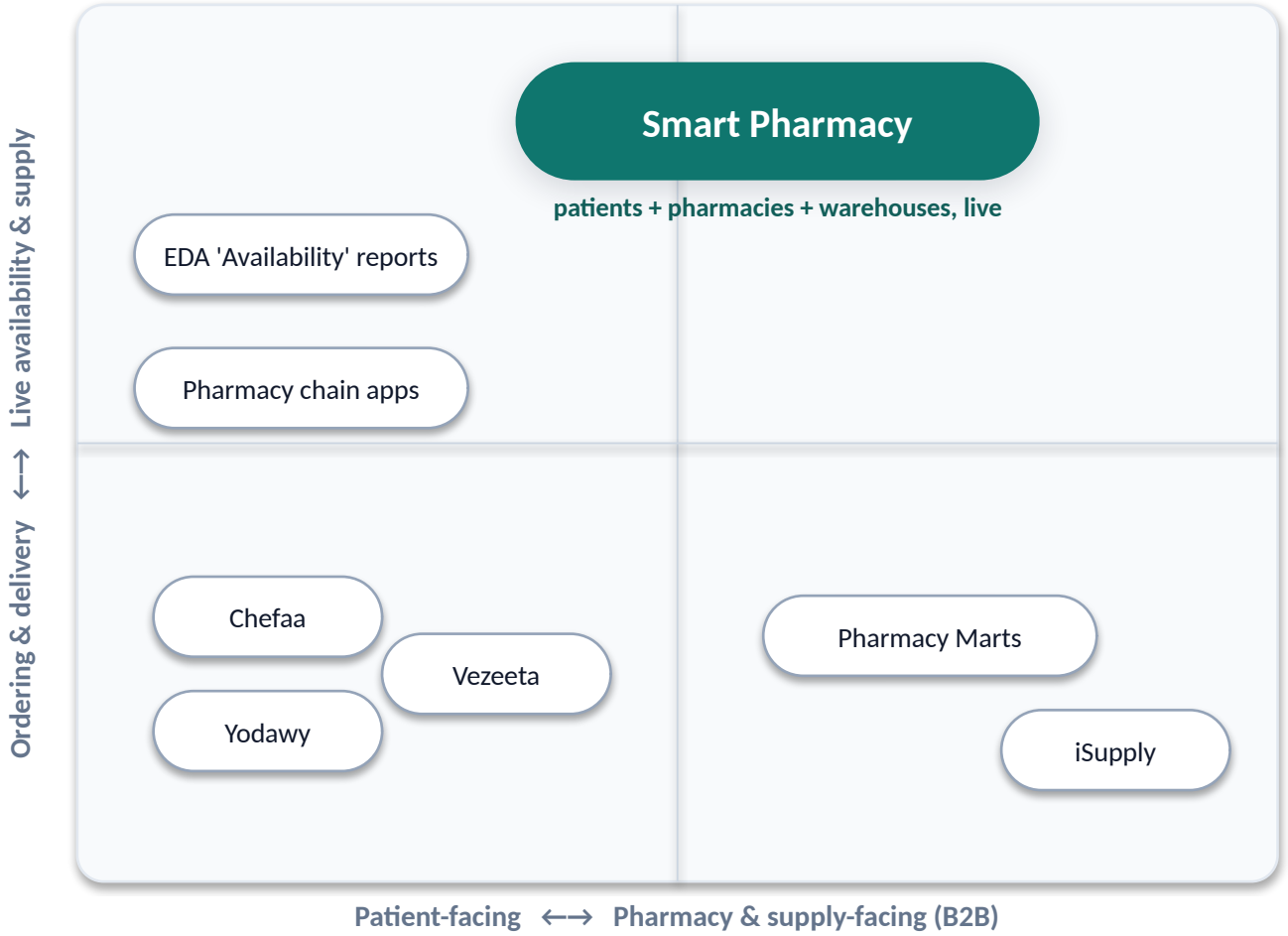


Synthetic load: 3,000 pharmacies · 90,000 stock lines · 2-vCPU sandbox · 109 requests/s at 10 concurrent users.

Sources: [18]

COMPETITION

Where we fit — and why we're different



What only we combine

1. Live, verified multi-pharmacy availability with freshness labels
2. Progressive 5 → 40 km search; Arabic-first; works without an account or GPS
3. Warehouse → pharmacy loop: incoming stock visible, receipts update stock
4. Auditable stock ledger — no overselling, every unit traceable
5. Anonymous demand signals (unmet searches) for suppliers & regulators

Positioning based on public company descriptions (2023–2026); competitors may offer features not shown — not a feature audit.

Sources: [12] [16] [17]

Free for patients. Valuable for the supply chain.

Pharmacy Pro (SaaS)

Free listing & stock updates; Pro adds ledger analytics, POS integration, team accounts and alerts.

Warehouse plan

Order & shipment management for distributors + a small fee per fulfilled order.

Demand insights

Aggregated, anonymised shortage & demand signals for distributors, manufacturers and authorities.

Proposed model — to be validated during the pilot.

Patients always free · no selling of personal data

Go-to-market

1. **Pilot · Ismailia**
   1.5M people [1] · 5–10 pharmacies + 1 warehouse · 8 weeks
2. **Suez Canal cities**
   Port Said · Suez · Ismailia governorate towns
3. **Greater Cairo**
   Partner with national distributors and pharmacy groups

Pilot KPIs
Search success rate · time-to-medicine · % fresh listings · request response time · shortage signal lead time

Sources: [1]

ROADMAP

Built in this hackathon — ready for a pilot

✓ Done · MVP

- Patient finder (AR / EN, RTL)
- Inventory ledger, CSV & barcode
- Orders, shipments & receiving
- Admin verification & audit logs
- 67 automated tests passing

2 Next 3 months

- Ismailia pilot with real pharmacies
- SMS / WhatsApp availability alerts
- Password reset & e-mail verification
- POS connectors
- Verified catalog import (EDA data)

3 6–12 months

- Demand forecasting & reorder suggestions
- Insurance & e-prescription partners
- Expansion to Greater Cairo
- Open API for health authorities
- Supplier performance scorecards

TEAM

The team behind Smart Pharmacy

AA

Ahmed Alsaholy

AS

Ahmed Shahen

KZ

Khaled Zakarya

ME

Mohamed Ekramy

FM

Fayez Mohamed

BUILT WITH

NestJS | PostgreSQL + PostGIS | Next.js | React | TypeScript | Tailwind CSS

Thank you — let's make every medicine findable. شكرًا

SOURCES

Data sources

[1] CAPMAS population clock via EgyptToday (19 Sep 2026): 109.5M residents; Ismailia 1.5M
https://www.egypttoday.com/Article/1/149579/Egypt%E2%80%99s-domestic-population-reaches-109-5-Million-CAPMAS

[2] Egyptian Pharmacists Syndicate records, in Expl. Res. Clin. Soc. Pharmacy (2023): 95,000 pharmacies; 313k pharmacists
https://www.sciencedirect.com/science/article/pii/S2667276623001658

[3] Egyptian Drug Authority via Daily News Egypt (3 Jun 2025): market EGP 216bn (2023) → 309bn (2024)
https://www.dailynewsegypt.com/2025/06/03/egypts-pharmaceutical-market-hits-egp-309bn-in-2024-marking-43-growth/

[4] Egyptian Drug Authority via Arab Finance (Jul 2026): EGP 422bn in 2025 (+37%), ~4bn packs
https://www.arabfinance.com/en/news/newdetails/exploring-egypt-rising-pharmaceutical-industry

[5] Middle East Monitor (30 Jul 2024): ~1,000 of 17,000 registered drugs short; Concor, Brufen, Cataflam; 30M+ hypertensive
https://www.middleeastmonitor.com/20240730-egyptians-face-illness-amid-a-serious-scarcity-of-medicines/

[6] Ahram Online (20 Aug 2024): shortages fell from ~1,000 to ~500 medicine types
https://english.ahram.org.eg/NewsContentP/1/530363/Egypt/Egypt-drug-shortage-crisis-Major-relief-around-the.aspx

[7] Business Monthly (23 Oct 2024): 800+ types short (FEI memo); 20–40% of trade names (CICMS)
https://businessmonthlyeg.com/pharmaceutical-shortages-worsen-in-egypt-how-government-and-pharmacists-respond/

[8] Zawia3 (25 Apr 2025): 1,471 branded drugs unavailable, 300 without alternatives (FEI Pharma Chamber, Mar 2025)
https://zawia3.com/en/psychiatric-medications/

[9] EnterpriseAM (3 Sep 2025): ~200 missing medicines in 2025; chronic & life-saving gaps persist
https://enterpriseam.com/egypt/2025/09/03/pharma-firms-push-for-price-hikes-as-shortages-and-black-market-resurface/

[10] Al Majalla (9 Dec 2025): shortages cut from 40% to under 5% of total supply
https://en.majalla.com/node/328622/business-economy/egypt%E2%80%99s-drug-price-increases-help-companies-not-patients

[11] The National (15 Feb 2024): Augmentin & chronic-drug shortages; ~11M with diabetes; black-market risk
https://www.thenationalnews.com/mena/egypt/2024/02/15/medicine-shortages-cairo-diabetes-drugs/

[12] Wamda (19 Mar 2024): ~30% of pharmacists' time on manual stock checks; 7,000+ products, 3,000 suppliers
https://www.wamda.com/2024/03/egypt-pharmacy-marts-secures-bridge-round-acasia-ventures

[13] Mada Masr (9 Mar 2021): law allows a pharmacist to own no more than two pharmacies
https://www.madamasr.com/en/2021/03/09/feature/economy/the-rise-and-fall-of-egypts-top-pharmaceutical-chain/

[14] Naos Solutions – Egyptians & Digital 2025 (DataReportal data): 96.3M internet users, 81.9%
https://naos-solutions.com/egyptians-and-digital-2025-report/

[15] IDSC via Zawya (20 Apr 2026): 92.6M mobile-internet subscriptions (Dec 2025)
https://www.zawya.com/en/business/technology-and-telecom/egypts-mobile-internet-users-hit-926mln-in-december-2025-idsc-hwrcfjnz

[16] Ahram Online: Egyptian Drug Authority 'Availability' service for reporting missing medicines
https://english.ahram.org.eg/UI/Front/Inner.aspx?NewsContentID=527162

[17] Competitor profiles – AccretiveEdge (Aug 2025: Yodawy, Chefaa, Vezeeta); Tracxn (Pharmacy Marts, iSupply)
https://accretiveedge.com/articles/healthtech-startups-middle-east/

[18] Smart Pharmacy prototype – team measurements, Oct 2026 (tests, latency on synthetic data, demo analytics)
Internal: docs/10-testing-and-quality.md and live demo database

Market figures as reported by the cited sources (retrieved Oct 2026). Screens and demo analytics use clearly labelled demo data from the prototype; performance figures are team measurements on synthetic data.